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Source: The Journal of the Torrey Botanical Society, 144(4) : 379-384

Published By: Torrey Botanical Society

URL: <https://doi.org/10.3159/TORREY-D-17-00010.1>

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Excluding deer increases the proportion of beech saplings originating from seed versus those of root sprout origin¹

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Abstract. American beech (*Fagus grandifolia* Ehrh.) regenerates both from seed and also clonally via root sprouts. Regenerating beech saplings often form dense and depauperate understories that cast deep shade and displace co-occurring species. The relative proportion of saplings that originate from seed versus root sprouts varies widely throughout the range of beech. Although the cause for that variation remains unclear, it may be linked to canopy or soil disturbances, the spread of beech bark disease (BBD), or overabundant deer. Here, we asked whether the long-term exclusion of deer and the absence of BBD would favor the regeneration of saplings (20–150 cm tall) of seed origin versus those of sprout origin. We addressed this question using deer exclosures (16 and 60+ yr old) and paired controls in one forest in Pennsylvania where BBD had caused major adult mortality and another where BBD was not present. We found that excluding deer significantly decreased the relative proportion of root sprouts from approximately 60% to approximately 25% in each forest, regardless of stand age, exclosure age, soil type, and presence or absence of BBD. Our findings provide evidence that deer, acting as herbivores, seed predators, agents of physical disturbance, or all of those simultaneously, create forest understories where root sprouts predominate. Although speculative, our results may apply to large regions because deer have been overabundant throughout much of the geographic range of American beech.

Key words: beech bark disease, deer, *Fagus grandifolia*, overbrowsing, root sprouts, root suckers

American beech (*Fagus grandifolia* Ehrh.) is a late-successional tree species that is widespread throughout the eastern deciduous forest. Regeneration of beech in the understory occurs both from seed and clonally from root sprouts (Peterson and Jones 1997). Beech saplings originating from seed often outnumber root sprouts, although that proportion varies widely among regions and root sprouts tends to increase under harsher climatic conditions or following canopy or soil disturbances (Held 1983, Morris *et al.* 2004). Beech bark disease (BBD) complex has caused high adult mortality of American beech throughout $\geq 50\%$ of its range (Evans *et al.* 2005, and citations therein). That mortality can contribute to a dense beech understory because beech seedlings and saplings typically survive or remain uninfected, thereby creating thickets that are composed of many individuals of beech; of which, some fraction are root sprouts (Ehrlich 1934; Houston 1975, 1994; Morin *et al.* 2007; Garnas *et al.* 2011; Cale *et al.*

2013). Surprisingly, the degree to which BBD stimulates root sprout production is unclear. It may be that canopy and soil disturbances (whether logging, wind, or BBD) all stimulate sprouting within the understory or root sprouts might be abundant before those disturbances. Jones and Raynal (1986) determined that BBD failed to stimulate root sprout production when the parent tree was still alive. In Michigan, Petrillo and Witter (2005) found a similar proportion of root sprouts in forest stands where BBD was present (17%) versus where it was absent (21%). To our knowledge, there are no studies that report the proportion of beech root sprouts before the arrival of BBD and then after periods of mortality caused by the pathogen. For example, although Houston (1975) reported the development of beech thickets after the death of adult beech, the proportion of beech saplings that were root sprouts versus those of seed origin was not reported.

Throughout much of the eastern deciduous forest biome, overabundant deer have caused substantial declines in understory diversity and favored a small subset of species that are both highly shade tolerant and also likely browse tolerant and that includes American beech (Kobe *et al.* 1995, Rooney 2001, Horsley *et al.* 2003, Côté *et al.* 2004, Long *et al.* 2007, Krueger *et al.* 2009, Nuttle *et al.* 2013). Beaudet and Messier (2008) demonstrated that beech root sprouts growing in the shaded understory have 20% higher

¹ We thank John Dzemyan and Alejandro Royo for assistance with data collection and other critical logistics, and Eric Griffin, Nathan Brouwer, and Alejandro Royo for helpful comments on earlier drafts of the manuscript.

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doi: 10.3159/TORREY-D-17-00010.1

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Received for publication August 31, 2016, and in revised form January 9, 2017; first published September 13, 2017.

survivorship than those from seedlings, and root sprouts can survive complete manual defoliation (55% survival), whereas seedlings cannot (0% survival). Nonetheless, it remains unknown the degree to which browsing (or seed predation on beechnuts by deer) might change the relative proportion of saplings that regenerate in the understory that arise from clonal root sprouts versus from seed. Kain *et al.* (2011) argued that root sprouts of beech would survive browsing better than seedlings and thus root sprouts should dominate where browsing pressure is high. However, beech has photosynthetic bark (e.g., Glass and Granet 1978), which may also provide some degree of browse tolerance (Kain *et al.* 2011). Thus, it is not clear the degree to which beech persists in the understory under heavy browsing because its bark is photosynthetic or because many saplings are root sprouts (with clonal connections to larger individuals) or both.

It is important to note that BBD and other canopy and soil disturbances can create or contribute to the formation of dense beech thickets whether they are of seed origin or of root sprout origin (Whitney 1984, Houston 2001, Hane 2003, Horsley *et al.* 2003, Banta *et al.* 2005, Long *et al.* 2007, Garnas *et al.* 2011, Cale *et al.* 2013, Nuttle *et al.* 2013). What remains unknown is the degree that the activity of deer may contribute to those thickets by creating an understory composed primarily of beech root sprouts before any disturbances. Thus, here we ask (a) whether browsing increases the relative proportion of root sprouts versus true seedlings, and (b) whether that proportion varies between an area where BBD has already caused substantial adult mortality versus an area where BBD is absent. If browsing causes an increase in the relative abundance of root sprouts when BBD spreads through a region, it may kill trees over a beech understory where root sprouts already predominate. Therefore, over-browsing and BBD together may create understory beech thickets dominated by clonal saplings. We address our questions using several relatively old deer exclosures (16 or 60+ yr old) in two mesophytic forests in Pennsylvania, one subject to BBD and one where BBD was absent.

Materials and Methods. **STUDY SITES.** We conducted this study in two forests in western and north-central Pennsylvania separated by ~200 km; beech was the dominant or codominant tree

species in both forests (Long *et al.* 2007, Kain *et al.* 2011). Tryon-Weber Woods is a 10-ha, mature beech-maple (*Acer* L. spp.) forest located in Crawford County, PA (41.5935°N, 80.3564°W; DeMars 1994, Long *et al.* 2007), and it currently lies just outside the western edge of the BBD killing front. The forest occurs over glacial till and is part of the beech-maple association (Braun 1950). This stand lies within a landscape matrix composed of closed-canopy, second-growth forests, old-fields, and agricultural habitats (Chips *et al.* 2014). This forest had no signs of BBD, such as canopy dieback or scale-induced alterations to the bark (Crawford County Natural Heritage Inventory 2008; W.P.C, personal observation). Deer densities throughout the region are ~13 deer/km², well above the densities seen before European colonization (~4 deer/km²; McCabe and McCabe 1984, 1997; Rosenberry *et al.* 2011). Browsing has strongly altered plant species composition and changed relationships between tree growth and mortality for understory saplings (Long *et al.* 2007, Pendergast *et al.* 2016).

The second forest was located on Pennsylvania State Game Lands 30 (41.6333°N, 78.3167°W) and occurs on unglaciated soil within the Allegheny high plateau region of Pennsylvania. This 70–80-yr-old, second-growth forest is part of the hemlock (*Tsuga* Carrière spp.)-northern hardwoods association (Whitney 1990) and is dominated by sugar maple (*Acer saccharum* Marshall), American beech, and black cherry (*Prunus serotina* Ehrh.) (Kain *et al.* 2011). Deer densities range from 8–10 deer/km² (Horsley *et al.* 2003, Rosenberry *et al.* 2011), and the surrounding landscape is primarily continuous forest. Within this forest and throughout the region, browsing has caused major shifts in the composition of both woody and herbaceous species (Rooney and Dress 1997, Banta *et al.* 2005, Comisky *et al.* 2005, Goetsch *et al.* 2011, Kain *et al.* 2011, Carson *et al.* 2014). BBD swept through this region beginning in the 1980s causing high mortality of adult beech (Mielke *et al.* 1986, Runkle 2007). Typically, BBD kills between 35% and 85% of adult beech, but that can vary based on slope position, competitors, geographic region, and tree size; smaller saplings (<10 cm diameter at breast height) typically survive or remain uninfected or both (Houston 1994; Evans *et al.* 2005, and citations therein).

EXCLOSURES. Within Tryon-Weber Woods (BBD absent), we sampled three 400-m² deer exclosures

(20 m × 20 m; height, 2.4 m; mesh size, 5 cm × 10 cm) and adjacent, randomly selected, paired control plots, which were established in 1996 (see Long *et al.* 2007, Pendergast *et al.* 2016). Exclosures and paired control plots were scattered haphazardly across the forest and were at least 50 m apart. Within the maple-beech-black cherry forest (BBD present), we sampled a single 0.4-ha exclosure and adjacent reference site (height, 2.1 m; mesh size, 15 cm × 20 cm). The exclosure has been maintained by the Pennsylvania Game Commission since the late 1940s and is > 60 yr old (Dzemyan 2009, Goetsch *et al.* 2011, Kain *et al.* 2011).

SAMPLING BEECH SAPLINGS. In May 2013, at Tryon-Weber Woods, we surveyed beech saplings 20–150 cm in height (stems within the reach of deer) for each of the three exclosures along a 2.5-m buffer that extended inside and outside the perimeter of each fence. We sampled that buffer area because we did not want to damage long-term census plots within the center of each exclosure and control plot.

In September 2013, within the maple-beech-black cherry forest, we surveyed beech stems 20–150 cm in height inside the fence and within an adjacent reference site with similar elevation, slope, and aspect. Within the exclosure and nearby reference site, we ran four, parallel, 60-m transects that were ~ 10 m apart. We recorded the origin of each stem (seed versus clonal) closest to the transect at eight randomly (stratified) located, 5-m intervals along each transect. We determined the reproductive origin of each stem within both forests by carefully and gently excavating the soil surrounding the base of each stem (Ward 1961, Held 1983, Jones and Raynal 1986, Nyland 2008). This exposed the lateral root and the associated root-sprout junction, or not, if it was of seed origin. Excavations were typically < 10 cm deep, and the determination was always unequivocal.

STATISTICAL ANALYSES. We calculated the percentage of seedlings of sprout origin as the number of root sprouts divided by the total number of stems (root sprouts + individuals of seed origin) inside the exclosures and in adjacent control plots. We then examined differences in the percentage of sprouts using a one-way ANOVA with SAS (PROC GLIMMIX, version 9.3.1, SAS Institute, Inc., Cary, NC). We fitted a β -distribution to the proportion (0–1) of root sprouts from each site to

satisfy assumptions of overdispersion (Kieschnick and McCullough 2003). To account for the heterogeneity associated with each paired control and exclosure, we included plot location as a random effect in our model for the Tryon-Weber Woods site. We did not account for plot in the model for the single 60+ yr-old exclosure but treated each transect as a pseudoreplicate (*sensu* Hurlbert 1984) and, thus, an independent unit of replication (following Goetsch *et al.* 2011, Kain *et al.* 2011). We fully acknowledge that this exclosure is unreplicated and thus our results must be considered with caution.

Results. Both at Tryon-Weber Woods (BBD absent) and ~ 200 km east, within a maple-beech-black cherry forest (BBD present), excluding deer for 16 and 60+ yr, respectively, led to understories in which ~ 20% of stems (20–150 cm in height) were root sprouts versus outside of exclosures where ~ 60% of stems were root sprouts (Fig. 1). We excavated a total of 134 and 64 saplings at Tryon-Weber Woods and the maple-beech-black cherry forest, respectively.

Discussion. We provide evidence that the presence of deer creates understories in which nearly two thirds of small beech saplings are clonal root sprouts; the opposite occurred inside exclosures where seedlings predominated (Fig. 1). This occurred in a forest in which BBD has been present for many years and caused high beech mortality and in a forest in which BBD remains absent. Our data suggest that when BBD moves through an area, if the region has a history of overbrowsing, most small beech saplings will be root sprouts. This may be true for larger beech saplings, depending on patterns of survivorship in root sprouts versus seedlings, as each move into larger size classes. Still, we only surveyed saplings between 20 cm and 150 cm tall. Nonetheless, we suggest that deer act as a strong biological filter (*sensu* Nuttle *et al.* 2013), creating understories in which root sprouts predominate even in larger sapling size classes (see Whitney 1984). Deer may increase the relative abundance of root sprouts via three mechanisms, the most obvious of which is that browsing, particularly in winter (Horsley *et al.* 2003), likely causes greater mortality of saplings of seed origin. However, deer also create soil disturbances via trampling and bedding, which may stimulate the production of root sprouts (Jones and Raynal 1986, 1987, 1988; Chips *et al.*

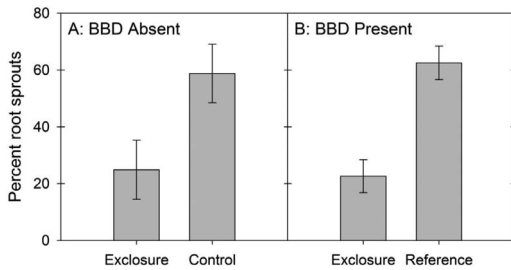


FIG. 1. (A) Tryon-Weber Woods. Mean (± 1 SE) percentage beech saplings (20–150 cm tall) of clonal origin found inside three 16-yr-old deer exclosures and in adjacent control plots in western Pennsylvania (d.f. = 1, 2; $F = 70.10$; $P = 0.014$). There were no signs of beech bark disease (BBD) in this forest. (B) State Game Lands 30. Mean (± 1 SE) percentage of beech saplings of clonal origin found inside a 60-yr-old deer exclosure and adjacent reference site within the Allegheny High Plateau region of Pennsylvania (d.f. = 1, 6; $F = 72.46$; $P = 0.004$). The BBD had been present at this site for ≥ 15 yr.

2014). Finally, deer eat beech seeds (Tubbs and Houston 1990), which would reduce the density of saplings of seed origin outside the exclosures. Regardless of the exact mechanism, overabundant deer have occurred across much of the eastern deciduous forest biome for decades (see introduction) and thus our findings may hold for large regions.

Our results suggest that deer may be reducing the genetic diversity of regenerating beech stands by creating understories in which root sprouts are genetically identical to their parents. If these root sprouts typically come from adults that have previously succumbed to BBD, that may create populations of saplings vulnerable to the disease (thus reducing the relative abundance of disease-resistant genotypes), and when these root sprouts mature, they too will likely become infected and die. Alternatively, in West Virginia, MacKenzie (2005) found that healthy trees, apparently resistant to BBD, produced more than five times as many root sprouts versus nearby trees that were already dead or showing signs of BBD. Thus, resistant, genetically identical, clonal root sprouts may serve as a reservoir and future source of resistant genotypes.

Our study comes with caveats. We sampled only one forest in which BBD was present and one in which the disease was absent. In addition, at one site, we sampled an unreplicated exclosure (albeit a 60-yr-old one) and a single adjacent reference site.

Therefore, our domain of generality is limited, and we cannot evaluate how broadly our results apply. Overall, our findings may best serve as a call for further research to evaluate how often, and to what degree, deer create forest understories in which beech saplings are predominantly root sprouts and the degree that may promote or delay the spread of resistant trees.

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